

Xynera Honeypot: Filesystem Mutation Analysis

Analysis and Resolution of the 'mkdir' and 'rmdir' Simulation Issues

Metadata	Detail
Report Focus	mkdir / rmdir Command Deception Routing and State Mutation
Affected System	xynera-honey (Deception Layer) & xynera-ai (Intelligence Layer)
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Resolution Status	Fixed and Tested successfully

1. Executive Summary

During standard interaction tests of the Xynera Honeypot system, it was discovered that directory creation (mkdir) and deletion (rmdir) commands executed by adversaries did not behave realistically. Instead of providing silent success or proper output, they generated 'command not found' errors, or failed to persist changes in subsequent directory listings (ls). This report details the root-cause analysis of these issues and explains the implemented solution that now enables dynamic virtual filesystem mutations.

2. Root Cause Identification

Issue A: Response Cleaning Logic (Command Routing)

In the intelligence layer (xynera-ai), commands that succeed silently (like mkdir or chmod) are matched against short-circuit rules and return an empty string (""). However, in the honeypot client's ai_client.py, the response cleaning helper mapped empty strings to None:
return text if text else None.

Because None signifies an AI backend request failure, the honeypot client's command router fell back to local routing. Since no local router entry existed for mkdir/rmdir, it hit the default command fallback, returning: [command]: command not found.

Issue B: Static, Read-Only Virtual Filesystem

Even if the empty response "" bypassed the cleaning error, the mock filesystem in fake_filesystem.py was entirely static. It was represented as a fixed module-level dictionary with no functions to update keys or list structures. Consequently, any created folders were never saved or reflected in subsequent commands like ls or cd, breaking the honeypot's simulation realism.

3. Implemented Solution

To resolve these challenges and preserve honeypot deception integrity, the following enhancements were implemented in the deception layer (xynera-honey):

1. Filesystem Mutator Helpers

Added robust in-memory mutation helper functions, `mkdir(session, path)` and `rmdir(session, path)`, to `fake_filesystem.py`. These functions parse absolute/relative paths, verify that parent directories exist, check for pre-existing files or subdirectories to prevent conflicts, and dynamically alter the in-memory filesystem registry.

2. Deception Layer Command Routing

Added direct handlers for `mkdir` and `rmdir` in the `command_router.py` file. The router splits arguments, filters command-line flags (such as `-p`), and runs the filesystem helpers for each directory target, returning empty strings on success or standard Linux system errors on failure.

4. Verification and Testing Results

A script was written to verify the changes. The script simulates an active SSH session and runs the command sequences. The output results show perfect alignment with standard Ubuntu terminal behavior:

Command Executed	Expected Output	Outcome / Verification
<code>mkdir abc</code>	[Empty response] ("")	Success. Directory 'abc' created.
<code>ls</code>	Lists 'abc' among others	Success. State persists in ls outputs.
<code>cd abc && pwd</code>	<code>/home/ubuntu/abc</code>	Success. Path resolution works correctly.
<code>rmdir abc</code>	<code>rmdir: failed to remove 'abc': Directory not empty</code>	Success. Safety checks block deletions of non-empty folders.
<code>rmdir abc/xyz && rmdir abc</code>	[Empty response] ("")	Success. Recursive clearing completes successfully.